
JPMorganChase ML Take Home Project

Project Report
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Chapter 1

Data Preprocessing and EDA

This section describes the data preprocessing and exploratory data analysis (EDA) conducted on the dataset. The dataset presents several challenges, including missing values, severe class imbalance, and a large number of categorical variables. Therefore, appropriate preprocessing steps, feature encoding methods, and model selection are crucial for improving the performance and interpretability of both the classification and clustering tasks.

1.1 Handling Missing Values

Some entries in the dataset are represented by the symbol “?”, which indicates missing values. I first converted these values into NaN so that the missing percentage of each column could be calculated. After examining the missing rates, I found that several columns had missing values for nearly 50% of their observations. These columns were removed because such a high missing rate could make imputation unreliable and introduce additional noise into the analysis.

Most of the remaining columns with missing values are categorical variables. Instead of dropping these rows, I replaced the missing values with a separate category labeled “missing”. This approach preserves the observations while allowing the model to learn whether missingness itself contains useful information.

For several numerical variables, including *wage per hour*, *capital gains*, *capital losses*, *dividends from stocks*, *number of persons worked for employer*, and *weeks worked in year*, I observed a high proportion of zero values. These zeros are meaningful rather than missing, since they may indicate that an individual did not receive wages, investment income, or work during the year. Therefore, instead of removing these variables, I created indicator variables to capture whether each original variable was equal to zero or nonzero. These indicators help the model distinguish between individuals with no activity in these income-related variables and those with positive values. Since these variables are closely related to income status, dropping them directly would likely remove useful predictive information for the classification task.

1.2 Exploratory Data Analysis and Feature Engineering

After preprocessing the missing values, I conducted exploratory data analysis (EDA) to better understand the relationship between demographic and employment-related variables and the income label. Since the target variable is highly imbalanced, I first examined the distribution of the income labels. The result shows that 93.79% of the dataset belong to the under-\$50K class, while only 6.21% belong to the over-\$50K class. This severe imbalance suggests that accuracy alone is not an appropriate evaluation metric for the classification task. Instead, metrics such as precision, recall, F1-score, ROC-AUC, and PR-AUC should be considered when evaluating model performance. Also, some techniques are applied to model training process which will be discussed later.

I then explored how income status varies across several important demographic and employment variables. For age, I first visualized the income label distribution by individual age (fig 2.1). The stacked bar chart shows that the over-\$50K group is concentrated mainly among working-age individuals, while very young individuals and older individuals have a much lower proportion of high-income observations. To make this pattern easier to interpret, I grouped age into three categories: 0–29, 29–59, and 60–90. This suggests that age is an important predictor of income status and may also be useful for customer segmentation.

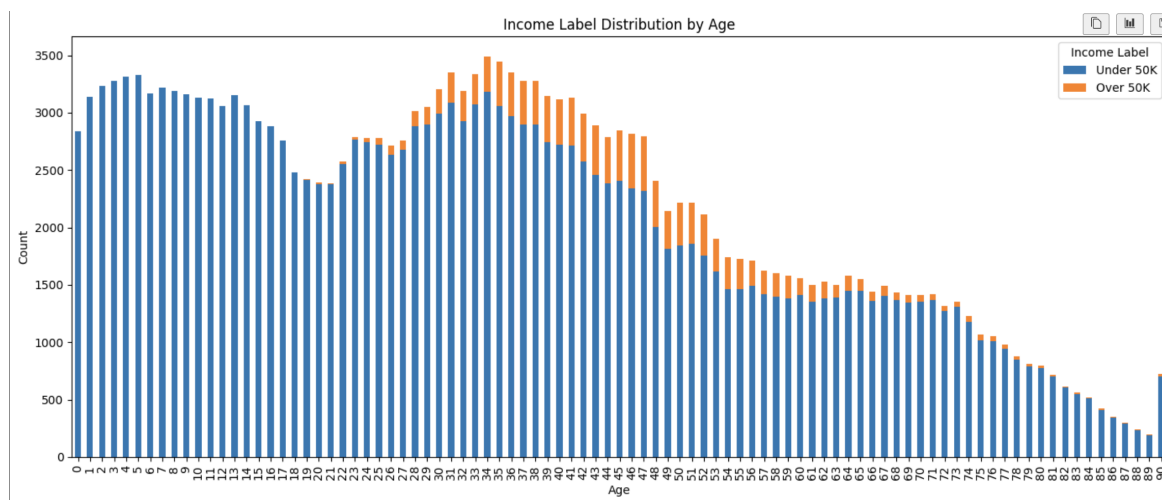


Figure 1.1: Income Label Distribution by Age

Next, I examined marital status. The distribution showed that individuals who are divorced or married with a civilian spouse present have a relatively stronger presence of the over-\$50K class compared with other marital groups. In contrast, categories such as never married, widowed, separated, or spouse absent are more heavily dominated by the under-\$50K class. Based on this pattern, I created a simplified feature called *marital_group*, which groups “Divorced” and “Married-civilian spouse present” into one category and

all other marital statuses into another category. This transformation reduces categorical complexity while preserving the main income-related pattern observed in the data.

Education showed a clear positive relationship with income: higher education levels generally had higher percentages of individuals earning over \$50K (Fig 2.2). Professional school and doctorate degree holders had the highest over-\$50K rates, followed by master's and bachelor's degree holders, while high school or lower education levels had much lower rates. Since education has a natural order, I encoded it as an ordinal variable so the model could capture the relationship between educational attainment and income.

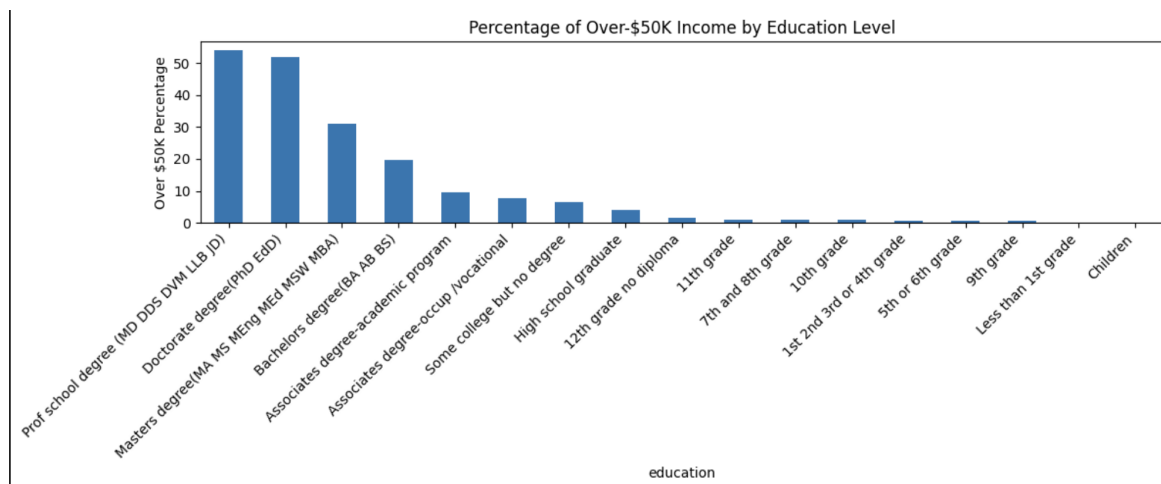


Figure 1.2: Percentage of Over 50 K income by education level

Employment-related variables also showed clear income differences. Executive, administrative, and managerial occupations had the highest over-\$50K rate, followed by professional specialty occupations, while sales, technicians, and protective services also showed relatively high rates. This suggests that occupation is an important feature because it reflects income potential and purchasing power.

Employment status and household role were also useful indicators. Full-time workers had a much higher over-\$50K rate than individuals not in the labor force, showing a strong connection between full-time employment and income. Household role showed a similar pattern, with householders having higher over-\$50K rates than children or other relatives, suggesting they are more likely to be income earners and decision-makers.

Based on the EDA results, I selected and engineered features for the modeling stage. Education was transformed into an ordinal variable, while most other categorical variables were one-hot encoded. I excluded duplicated or less interpretable variables when necessary, such as using broader occupation and industry categories instead of detailed recode variables. I also removed variables such as *weight* from the feature matrix, since it represents the sampling weight rather than an individual characteristic.

Chapter 2

Classification

In this section, I evaluate three different models for the income classification task: Logistic Regression, HistGradientBoosting, and a Multilayer Perceptron. These models were selected to compare the performance of a linear baseline model, a tree-based ensemble model, and a neural network model. The results show that the tree-based model performs better than both the linear model and the deep learning model, suggesting that HistGradientBoosting is more effective at capturing the nonlinear relationships between demographic, employment-related, and financial variables in this dataset.

2.1 Baseline Model: Logistic Regression

I first trained a Logistic Regression model as the baseline classifier because it is simple, interpretable, and provides a useful reference point for comparing more complex models. Since the dataset is highly imbalanced, with only 6.21% of observations belonging to the over-\$50K class, I used `class_weight="balanced"` to give more importance to the minority class during training.

The dataset was split into training, development, and test sets using stratified sampling to preserve the original label distribution. I also applied standard scaling because Logistic Regression is sensitive to feature scale. On the development set, the initial model achieved high recall for the over-\$50K class but relatively low precision, meaning it identified many true high-income individuals but also produced many false positives.

To improve the balance between precision and recall, I tuned the probability threshold on the development set. The best threshold based on class-1 F1-score was 0.85. Using this threshold, the final Logistic Regression model achieved a test accuracy of 94.25%, with a precision of 0.53, recall of 0.61, and F1-score of 0.57 for the over-\$50K class. The model also achieved a PR-AUC of 0.602, which is much higher than the original positive class rate of 6.21%.

Overall, Logistic Regression provides a strong and interpretable baseline. However, because it is a linear model, it may not fully capture complex nonlinear relationships in the dataset, so more flexible models are also explored.

2.2 HistGradientBoosting Model

After training the Logistic Regression baseline, I trained a HistGradientBoosting model to capture more complex and nonlinear relationships in the dataset. Compared with Logistic Regression, gradient boosting is more flexible because it builds an ensemble of decision trees and can model interactions among demographic, employment-related, and financial variables. This makes it suitable for tabular data with mixed numerical and categorical features.

Because the target variable is highly imbalanced, I used balanced sample weights during training. The over-\$50K class only accounts for 6.21% of the dataset, so without adjustment the model may be biased toward predicting the majority under-\$50K class. By applying `compute_sample_weight` with `class_weight="balanced"`, errors on the minority class were given higher importance during model training. This helped the model identify more high-income individuals instead of simply optimizing for overall accuracy.

The initial HistGradientBoosting model achieved strong performance on the development set, with a ROC-AUC of approximately 0.955. However, using the default classification threshold still produced a tradeoff between precision and recall for the minority class. Therefore, I tuned the probability threshold from 0.10 to 0.90 and selected the threshold that maximized the F1-score for the over-\$50K class. The best threshold was 0.85, which provided a better balance between precision and recall. At this threshold, the model achieved a development accuracy of approximately 94.80%, with a precision of 0.57, recall of 0.64, and F1-score of 0.60 for the over-\$50K class.

To make the model simpler and more interpretable, I performed feature selection using permutation importance on the development set. This avoids using the test set during model selection and keeps the test set reserved for final evaluation. The permutation importance results showed that age, education, weeks worked in year, capital gains, tax filer status, occupation, dividends from stocks, and number of persons worked for employer were among the most important predictors. Based on these results, I retrained a simplified HistGradientBoosting model using only the top 30 most important features. This reduced model preserved most of the predictive performance while making the final model easier to interpret and explain from a business perspective.

The final simplified HistGradientBoosting model achieved a test accuracy of approximately 94.61% and a ROC-AUC of approximately 0.950. For the over-\$50K class, the model achieved a precision of 0.56, recall of 0.65, and F1-score of 0.60. Compared with the Logistic Regression baseline, the HistGradientBoosting model improved the minority-class F1-score and PR-AUC, showing that the tree-based model was better at capturing the nonlinear patterns associated with higher income. Therefore, HistGradientBoosting was selected as the strongest classification model in this project.

2.3 Neural Network Model

In addition to Logistic Regression and HistGradientBoosting, I also trained a neural network model for the income classification task. The purpose of this model was to test whether a deep learning approach could capture complex nonlinear relationships in the processed feature space. Before training, all input features were standardized using *StandardScaler*, since neural networks are sensitive to feature scale. The processed data were then converted into PyTorch tensors and loaded using mini-batch data loaders.

The neural network was implemented as a multilayer perceptron with three hidden layers. The model used fully connected layers with hidden dimensions of 256, 128, and 64, followed by Tanh activation functions, batch normalization, and dropout. Batch normalization was used to stabilize training, while dropout was added to reduce overfitting. The final output layer produced one logit, which was converted into a probability using the sigmoid function during evaluation.

To address the severe class imbalance in the dataset, I used *BCEWithLogitsLoss* with a positive class weight. The positive class weight was calculated as the ratio between the number of negative and positive training examples, giving higher penalty to mistakes on the minority over-\$50K class. The model was trained using the Adam optimizer with weight decay for regularization. During training, the development-set ROC-AUC was monitored, and the best model state was selected based on development performance.

After training, I tuned the classification threshold on the development set to improve the balance between precision and recall for the over-\$50K class. The final selected threshold was 0.90. On the test set, the neural network achieved an accuracy of 95.13%, ROC-AUC of 0.943, and PR-AUC of 0.609. For the over-\$50K class, the model achieved a precision of 0.65, recall of 0.46, and F1-score of 0.54. This indicates that the neural network was relatively conservative when predicting high-income individuals: it produced fewer false positives, but it also missed more true over-\$50K cases compared with the HistGradientBoosting model.

Overall, the neural network achieved strong overall accuracy and reasonable ranking performance. However, its minority-class F1-score was lower than the HistGradientBoosting model. This suggests that, for this tabular dataset, the tree-based model was more effective than the deep learning model at capturing the most useful patterns for income classification.

2.4 Model Comparison

Table 2.1 compares the performance of the three classification models. Since the dataset is highly imbalanced and the business goal is to identify individuals earning over \$50K, overall accuracy is not the most important metric. A model can achieve high accuracy by mostly predicting the majority under-\$50K class. Therefore, I focus more on precision, recall, and F1-score for the over-\$50K class, especially F1-score because it balances precision

and recall.

Among the three models, HistGradientBoosting achieved the best balance for the minority class, with the highest class-1 F1-score of 0.60 and recall of 0.65. Although the neural network had the highest overall accuracy, its recall for the over-\$50K class was lower, meaning it missed more potential high-income individuals. Therefore, HistGradientBoosting was selected as the final classification model.

Table 2.1: Comparison of Classification Model Performance

Model	Accuracy	ROC-AUC	PR-AUC	Precision ₁	Recall ₁	F1 ₁
Logistic Regression	0.943	0.941	0.602	0.53	0.61	0.57
HistGradientBoosting	0.946	0.950	0.653	0.56	0.65	0.60
Neural Network	0.951	0.943	0.609	0.65	0.46	0.54

Chapter 3

Clustering

This chapter presents the process of developing a segmentation model to identify different potential customer groups in the dataset. The main method I choose for this task is K-means clustering, which groups similar individuals based on selected features and helps reveal meaningful patterns within the data.

3.1 Feature Selection

For the segmentation model, the main goal is to identify meaningful customer groups and explain the resulting clusters using interpretable patterns in the data. Therefore, I prioritized variables that are easier to understand from a business perspective, such as age, education, occupation, employment status, household role, and income-related financial variables. Sensitive demographic features, including sex, race, and Hispanic origin, were excluded from the segmentation model for ethical marketing considerations. This helps reduce the risk of creating customer segments based on protected or sensitive demographic characteristics.

For preprocessing, numerical variables were standardized because K-means clustering is distance-based and sensitive to feature scales. Without scaling, variables with larger numerical ranges could dominate the distance calculation and have an unfair influence on the clustering result. Categorical variables were transformed using one-hot encoding so that they could be included in the clustering algorithm. Since one-hot encoding created a high-dimensional feature space, I applied TruncatedSVD to reduce dimensionality before clustering. This step helps improve computational efficiency and reduces noise while still preserving the main structure of the processed features.

3.2 Model Training Process and Cluster Number Selection

After preprocessing and dimensionality reduction, I trained K-Means models with different numbers of clusters. I evaluated the clustering results using both silhouette score (Fig 4.1) and the elbow method. The silhouette score measures how well each observation fits within its assigned cluster, while the elbow method uses inertia, or within-cluster sum of squares, to measure cluster compactness.

	k	silhouette_score	inertia
0	2	0.294848	1.847350e+06
1	3	0.293701	1.602949e+06
2	4	0.314835	1.409281e+06
3	5	0.346426	1.226643e+06
4	6	0.366139	1.070488e+06
5	7	0.384062	9.313622e+05
6	8	0.406464	8.052608e+05

Figure 3.1: Silhouette score of different number of clusters

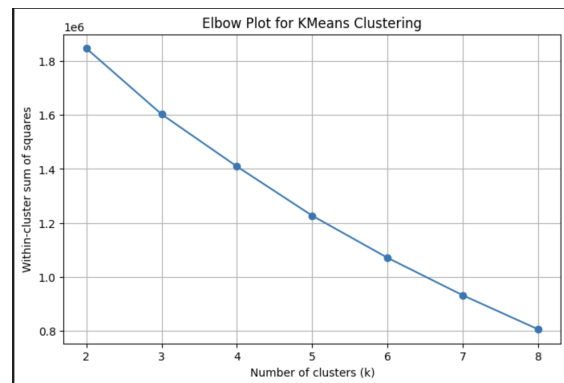


Figure 3.2: Elbow plot

Although the silhouette score increased as the number of clusters increased, I did not choose the final k based only on this metric. For business segmentation, the clusters also need to be meaningful and easy to interpret. Therefore, I compared the cluster profiles for $k = 3, 4, 5$, and 6.

The $k = 3$ model produced broad and understandable groups, but it was too general and missed an important high-value segment. The $k = 5$ and $k = 6$ models created more detailed groups, but some clusters were very similar or mainly separated by wage patterns, which made them less useful for interpretation. In contrast, $k = 4$ provided a good balance between detail and interpretability. It preserved the major population groups while also identifying a smaller, higher-income and financially active working segment.

Therefore, I selected $k = 4$ as the final number of clusters. Even though larger k values

had higher silhouette scores, $k = 4$ avoided unnecessary over-segmentation and produced more meaningful customer groups for business analysis.

3.3 Clusters Explanation

The final four-cluster model separates the population into four meaningful business segments (Table 4.1). Segment 0 represents an older low-employment group with low wage, few weeks worked, and a low over-\$50K rate, suggesting limited purchasing power and stronger interest in value-based products. Segment 1 mainly consists of children or dependent individuals, so this group is less useful for direct income-based marketing but may be relevant for household or family-oriented campaigns. Segment 2 is the largest and most important general consumer segment, consisting mainly of working-age adults with stable employment and moderate income potential. This group can be targeted with everyday retail products, loyalty programs, and convenience-based promotions. Segment 3 is a smaller but high-value segment with the highest over-\$50K rate, higher education, stronger work activity, and a higher dividend rate. This group is likely to have stronger purchasing power and can be targeted with premium products, personalized offers, and higher-value marketing campaigns.

Table 3.1: Business Interpretation of Final Customer Segments

Segment	Business Group Name	Main Characteristics	Marketing Insight
0	Older Low-Employment / Low-Income Group	Average age around 58, low weeks worked, low wage, and low over-\$50K rate. Most individuals are not in the labor force or not in universe.	This group likely has lower purchasing power. Marketing should focus on basic needs, discounts, household essentials, and value-oriented products.
1	Children / Dependent Group	Very young average age, very low education level, almost no work activity, and nearly zero over-\$50K rate. Most individuals are children or dependents.	This group should not be targeted directly for income-based marketing. Retail strategies should focus on parents or households, such as family products, children's items, school supplies, and household bundles.
2	Main Working Consumer Group	Largest segment, with working-age adults, high weeks worked, moderate education, and stable employment. Common industries include retail, manufacturing, education, and finance/real estate.	This is the core retail customer base. Marketing can focus on everyday consumer goods, convenience, work-related products, loyalty programs, and mid-range products.
3	High-Value Financially Active Working Group	Smallest segment but highest over-\$50K rate. This group has higher education, stable work activity, higher dividend rate, and more professional, executive, managerial, and sales occupations.	This is the most valuable premium segment. Marketing can focus on higher-end products, premium memberships, home goods, technology products, personalized offers, and higher-value campaigns.

Chapter 4

Conclusion

This project developed both a supervised classification model and an unsupervised customer segmentation model to support the retail client's marketing decisions. The classification task predicted whether an individual earns over \$50K, while the clustering task identified meaningful customer groups. Before modeling, I addressed missing values, skewed numerical variables, many categorical features, and severe class imbalance, which improved both performance and interpretability. For classification, I compared Logistic Regression, HistGradientBoosting, and a Neural Network. Since only 6.21% of observations belonged to the over-\$50K class, I focused on precision, recall, F1-score, ROC-AUC, and PR-AUC rather than accuracy alone. Although the Neural Network had the highest accuracy, HistGradientBoosting achieved the best balance between precision and recall for the minority class, so it was selected as the final model. For segmentation, I used K-means clustering with demographic, employment, household, and income-related variables, while excluding sensitive features such as sex, race, and Hispanic origin for ethical reasons. After preprocessing, scaling, encoding, and dimensionality reduction, four clusters were selected based on evaluation results and business interpretability. These clusters represented an older low-employment group, a children/dependent group, a main working consumer group, and a high-value financially active working group. Overall, combining classification and segmentation provides stronger business insight than using either method alone. The classification model identifies likely high-income customers, while the segmentation model explains broader customer profiles and supports targeted marketing strategies for different groups.

Reference:

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